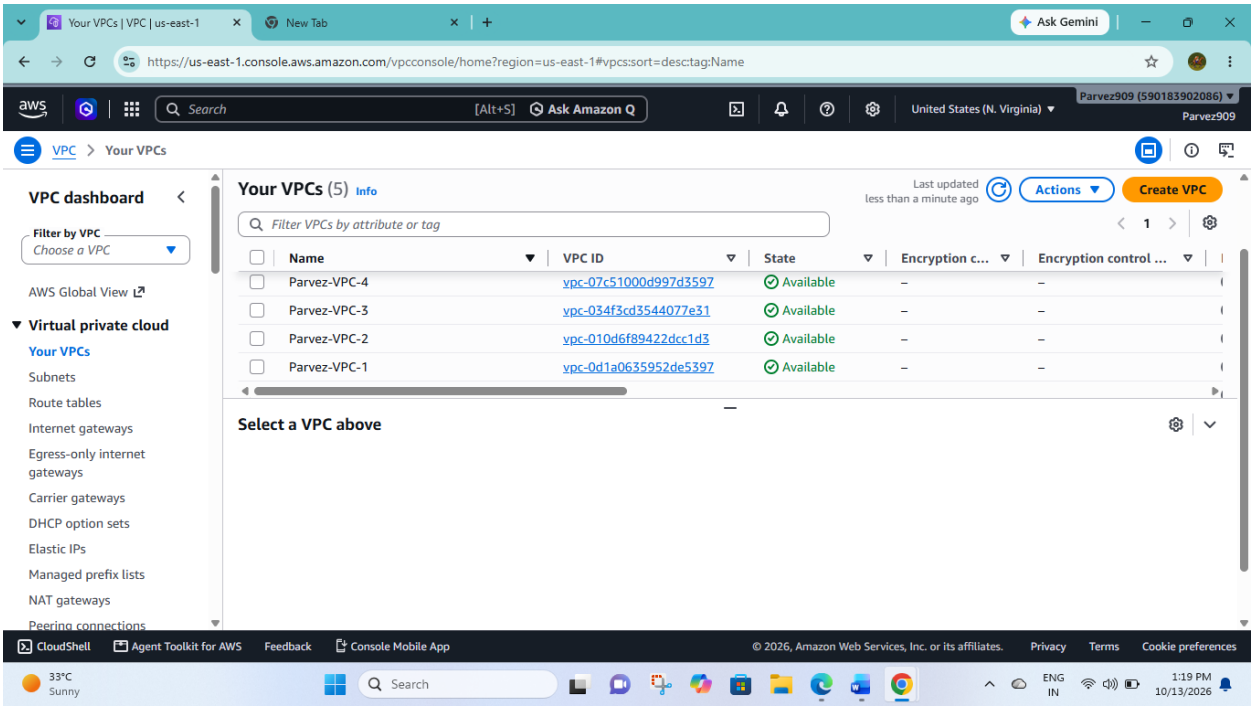
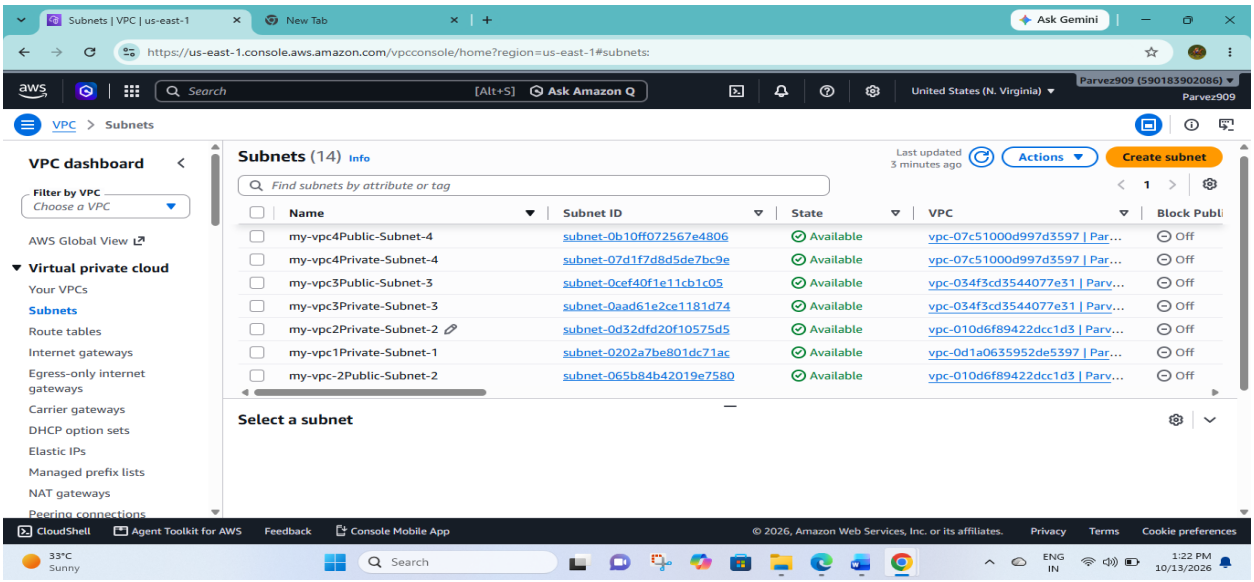
VPC Transit Gateway & VPC Endpoint challenges

1. Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs

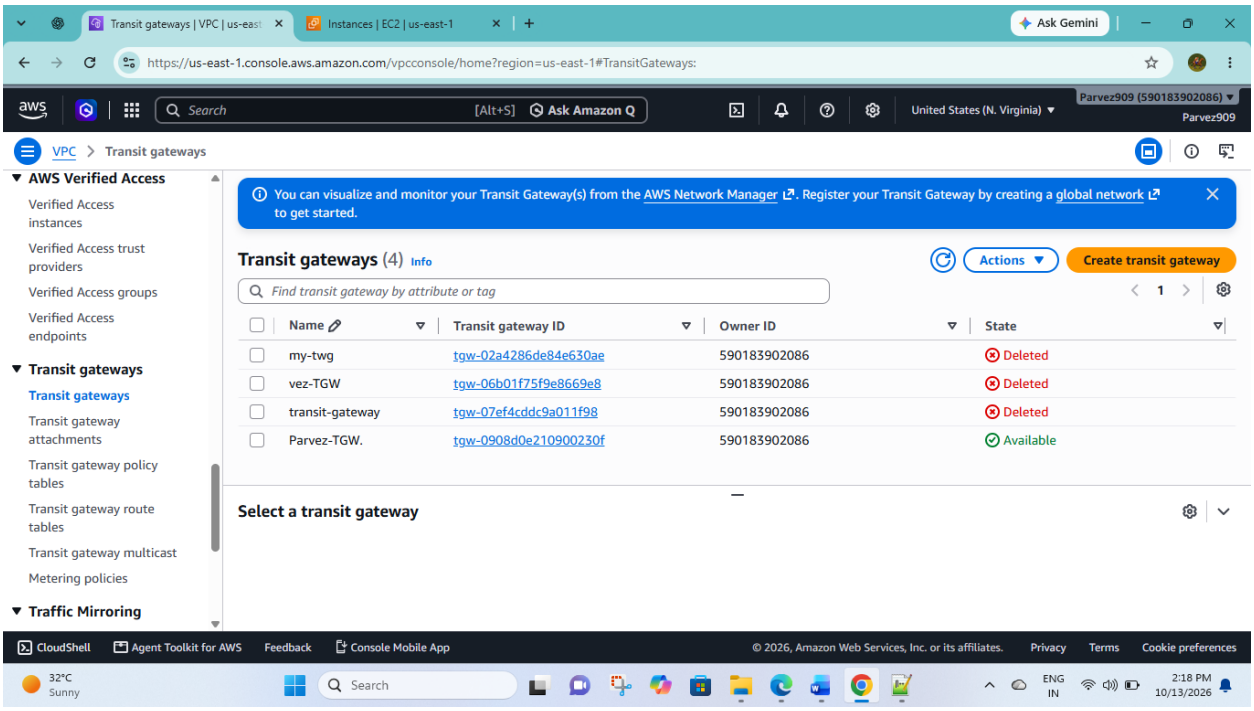
1.successfully created 4 vpc.



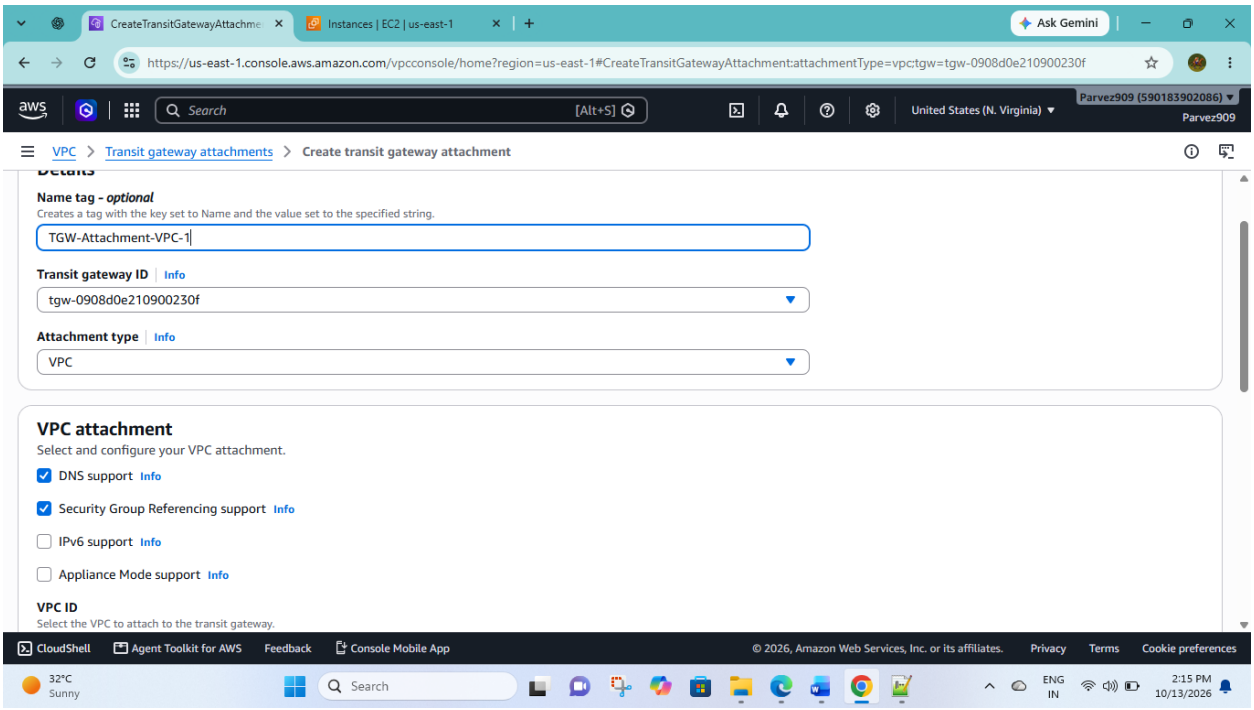
2.sucessfully created 4 subnets public and private



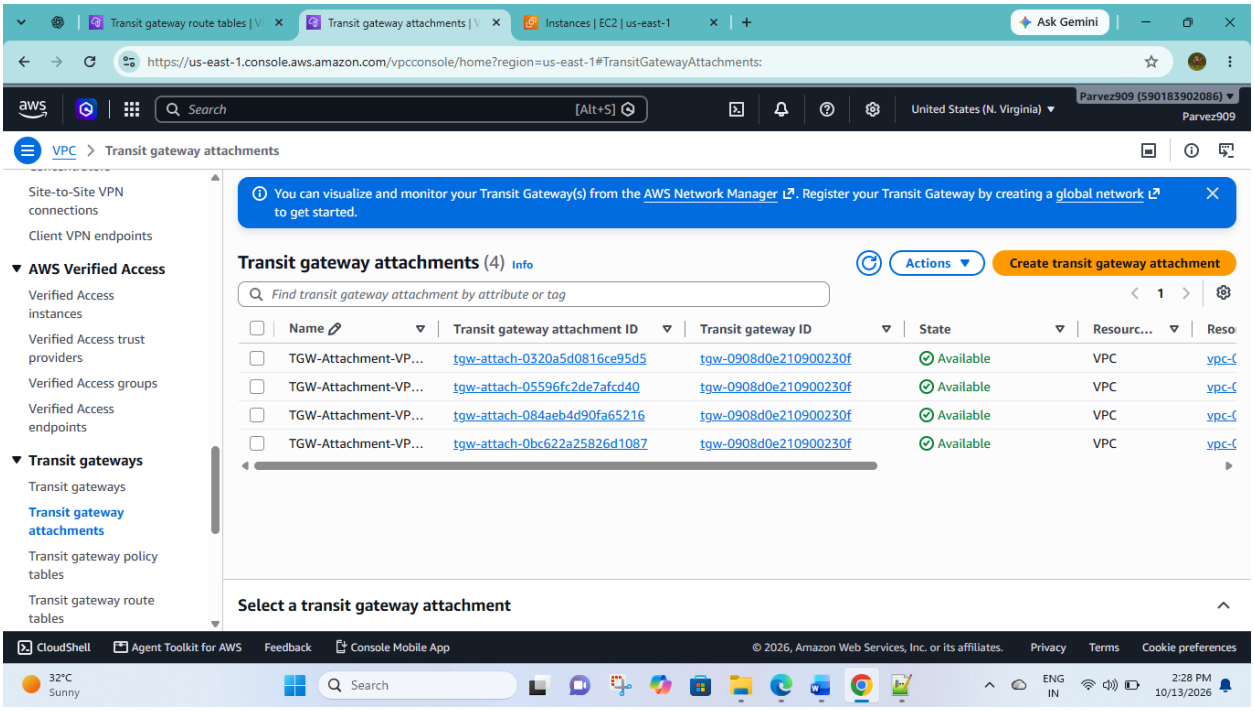
3. Create a Transit Gateway



4. Create four VPC attachments

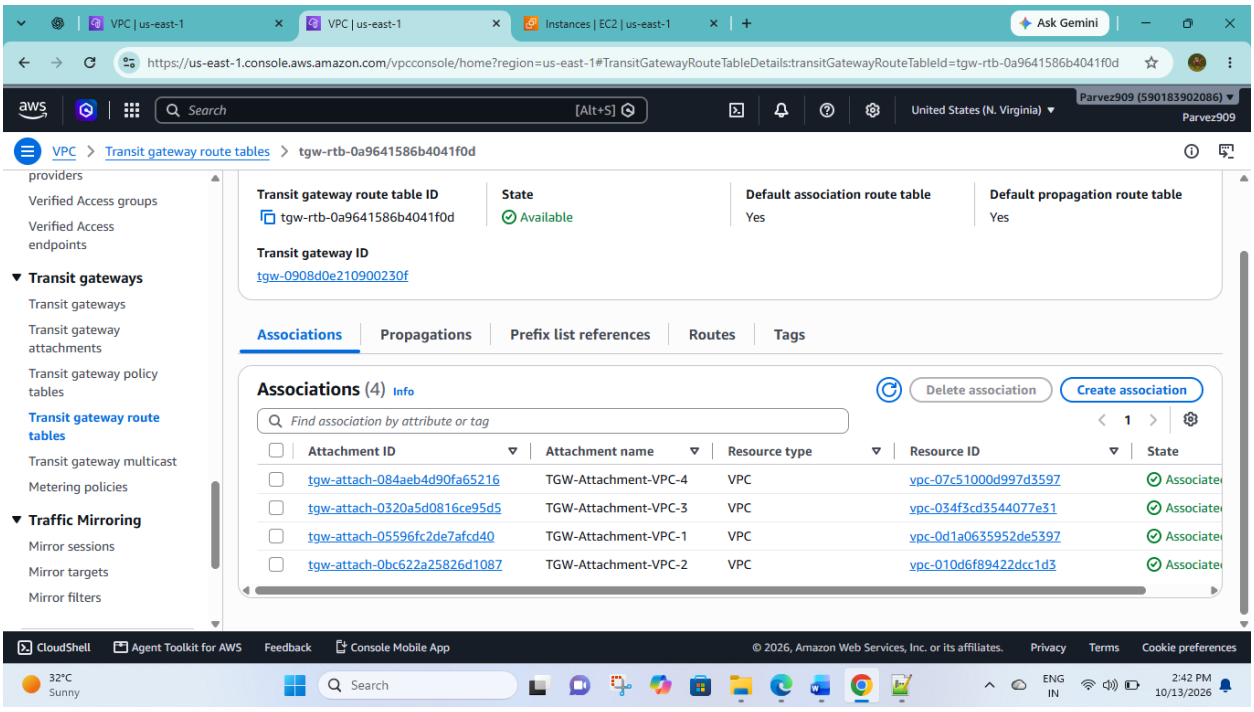


Repeat for VPC-2, VPC-3, and VPC-4.



5.Configure the Transit Gateway route table

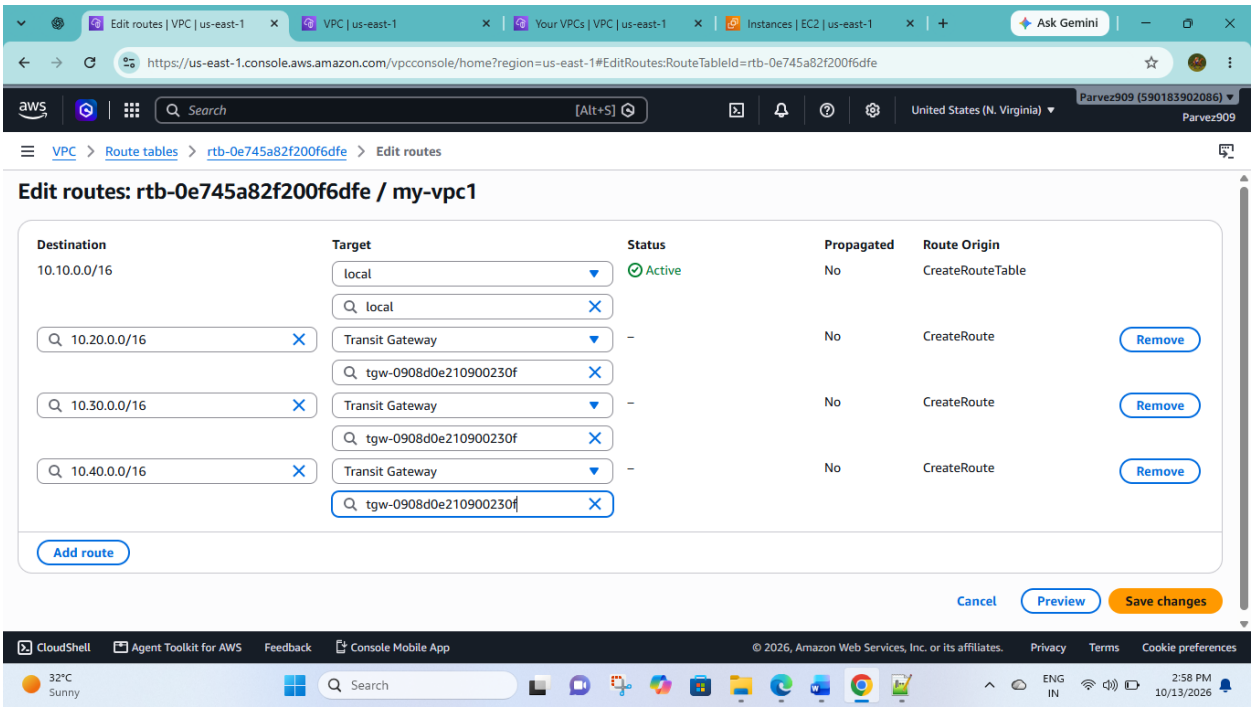
By default one route table will we have so just click we will see the all vpc attachments.



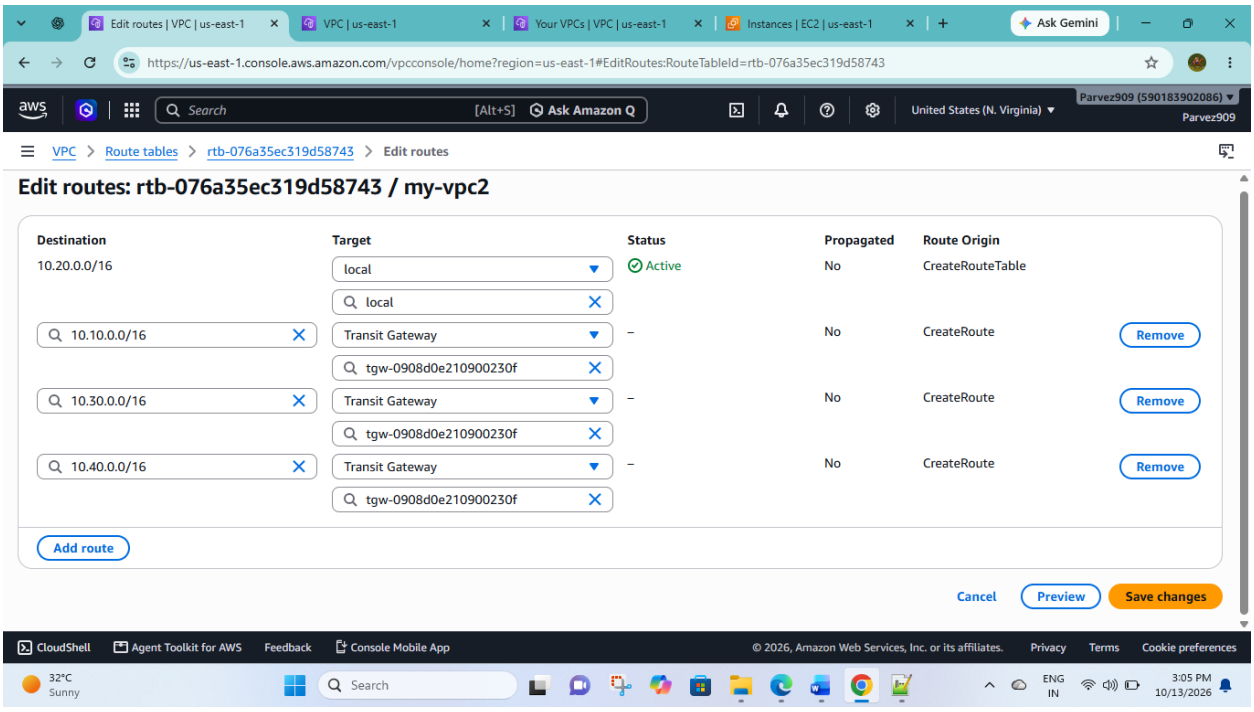
6.Update the VPC route tables

For each VPC, open Route Tables → select the route table used by your private subnet → Routes → Edit routes.

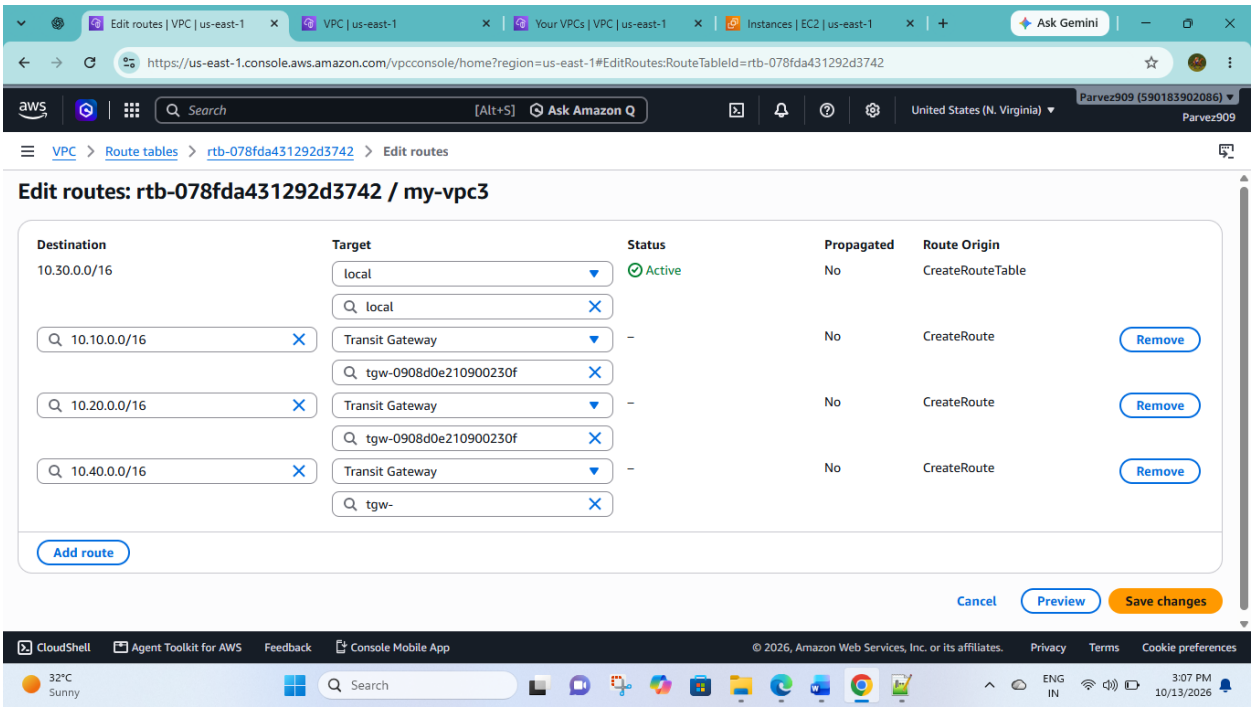
Vpc-1



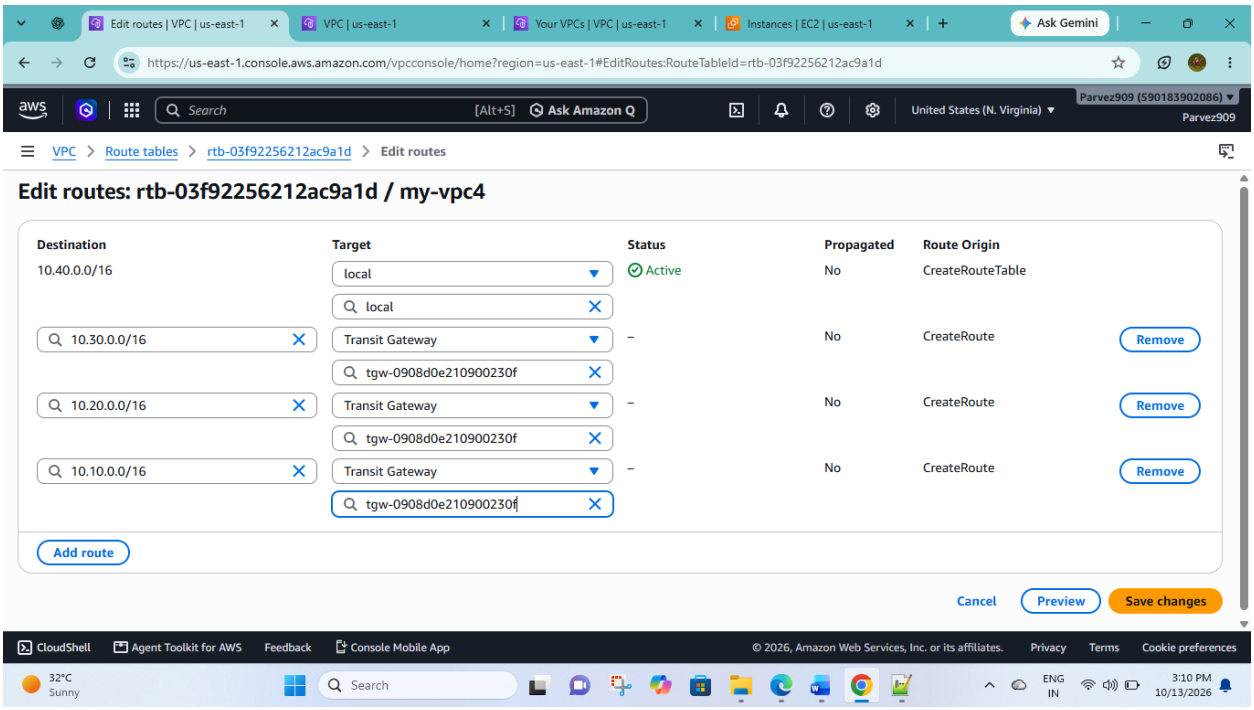
Vpc-2

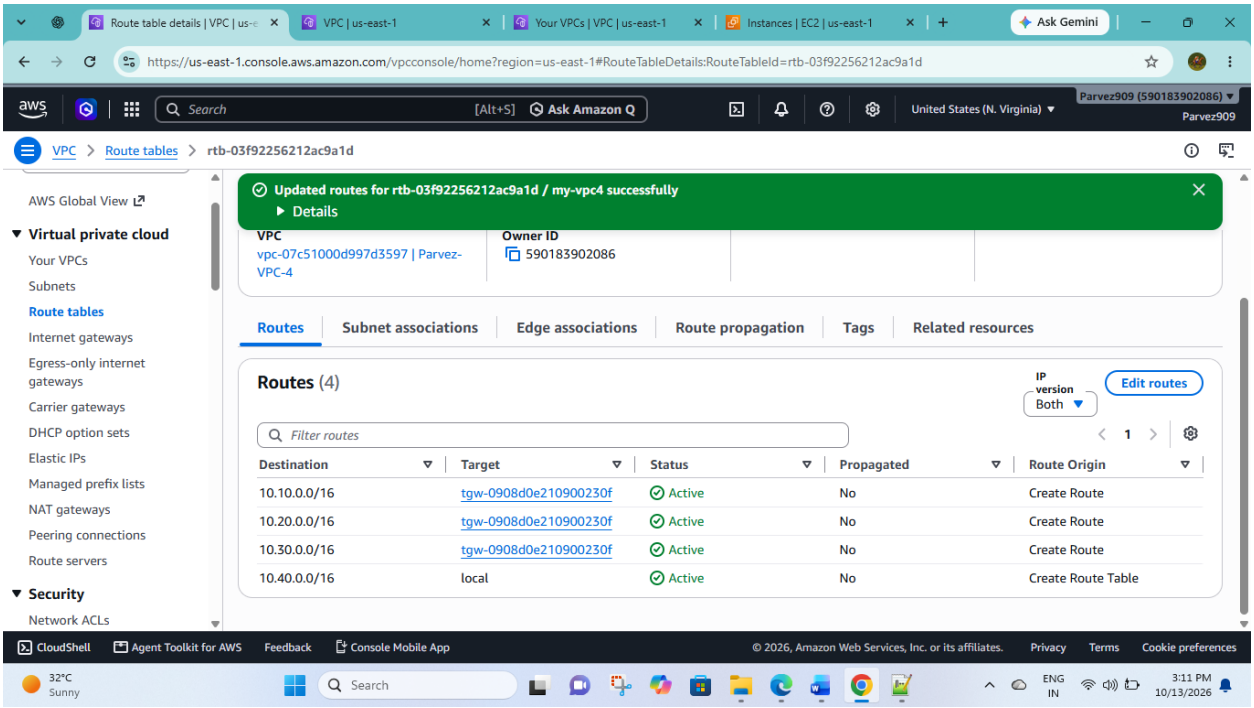


Vpc-3


Edit routes | VPC | us-east-1
VPC | us-east-1
Your VPCs | VPC | us-east-1
Instances | EC2 | us-east-1
Ask Gemini
https://us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#EditRoutes:RouteTableId=rtb-078fda431292d3742
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
VPC > Route tables > rtb-078fda431292d3742 > Edit routes
Edit routes: rtb-078fda431292d3742 / my-vpc3
Destination
Target
Status
Propagated
Route Origin
10.30.0.0/16
local
Active
No
CreateRouteTable
local
10.10.0.0/16
Transit Gateway
–
No
CreateRoute
Remove
tgw-0908d0e210900230f
10.20.0.0/16
Transit Gateway
–
No
CreateRoute
Remove
tgw-0908d0e210900230f
10.40.0.0/16
Transit Gateway
–
No
CreateRoute
Remove
tgw-
Add route
Cancel
Preview
Save changes
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
32°C
Sunny
Search
ENG
IN
3:07 PM
10/13/2026

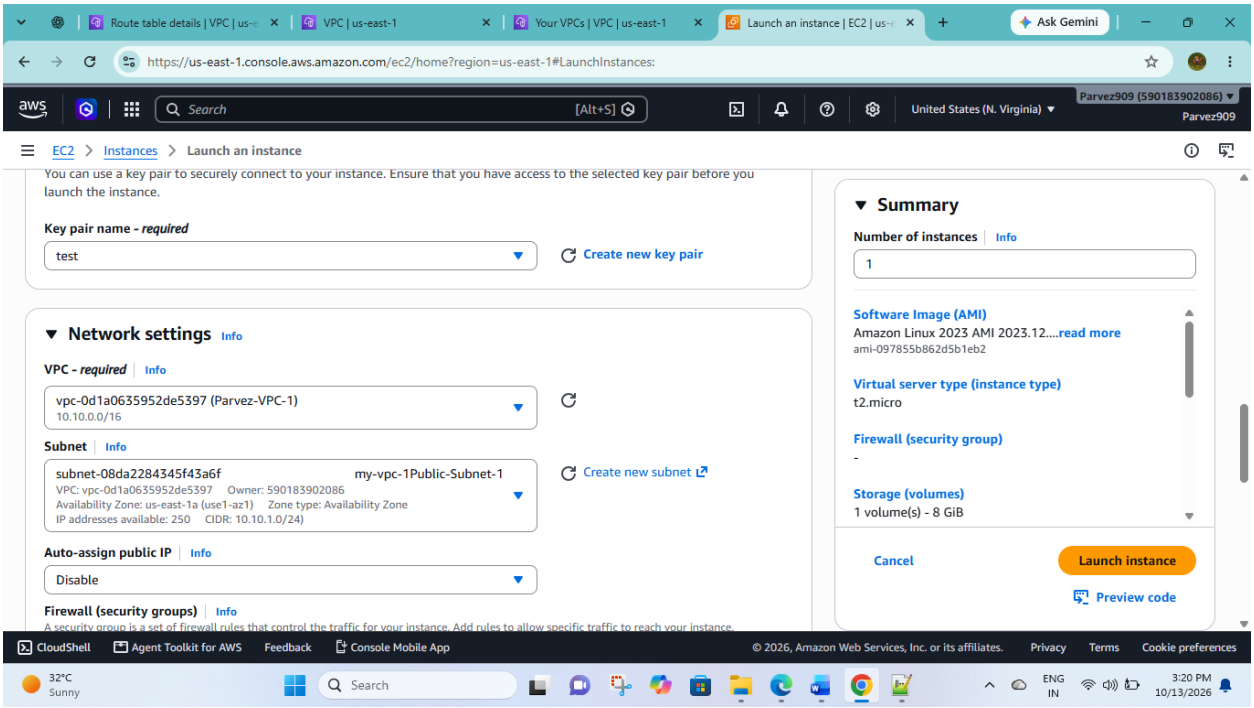
Vpc-4

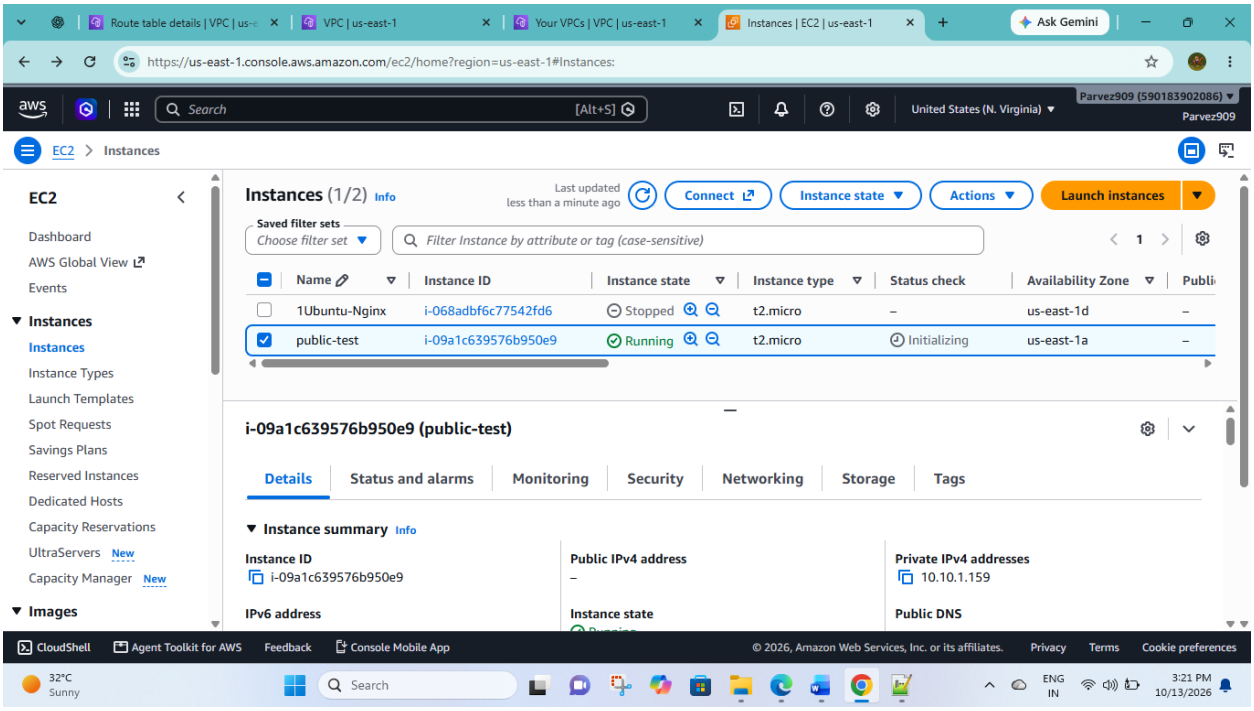

Edit routes | VPC | us-east-1
VPC | us-east-1
Your VPCs | VPC | us-east-1
Instances | EC2 | us-east-1
Ask Gemini
https://us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#EditRoutes:RouteTableId=rtb-03f92256212ac9a1d
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
VPC > Route tables > rtb-03f92256212ac9a1d > Edit routes
Edit routes: rtb-03f92256212ac9a1d / my-vpc4
Destination
Target
Status
Propagated
Route Origin
10.40.0.0/16
local
Active
No
CreateRouteTable
local
10.30.0.0/16
Transit Gateway
–
No
CreateRoute
Remove
tgw-0908d0e210900230f
10.20.0.0/16
Transit Gateway
–
No
CreateRoute
Remove
tgw-0908d0e210900230f
10.10.0.0/16
Transit Gateway
–
No
CreateRoute
Remove
tgw-0908d0e210900230f
Add route
Cancel
Preview
Save changes
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
32°C
Sunny
Search
ENG
IN
3:10 PM
10/13/2026

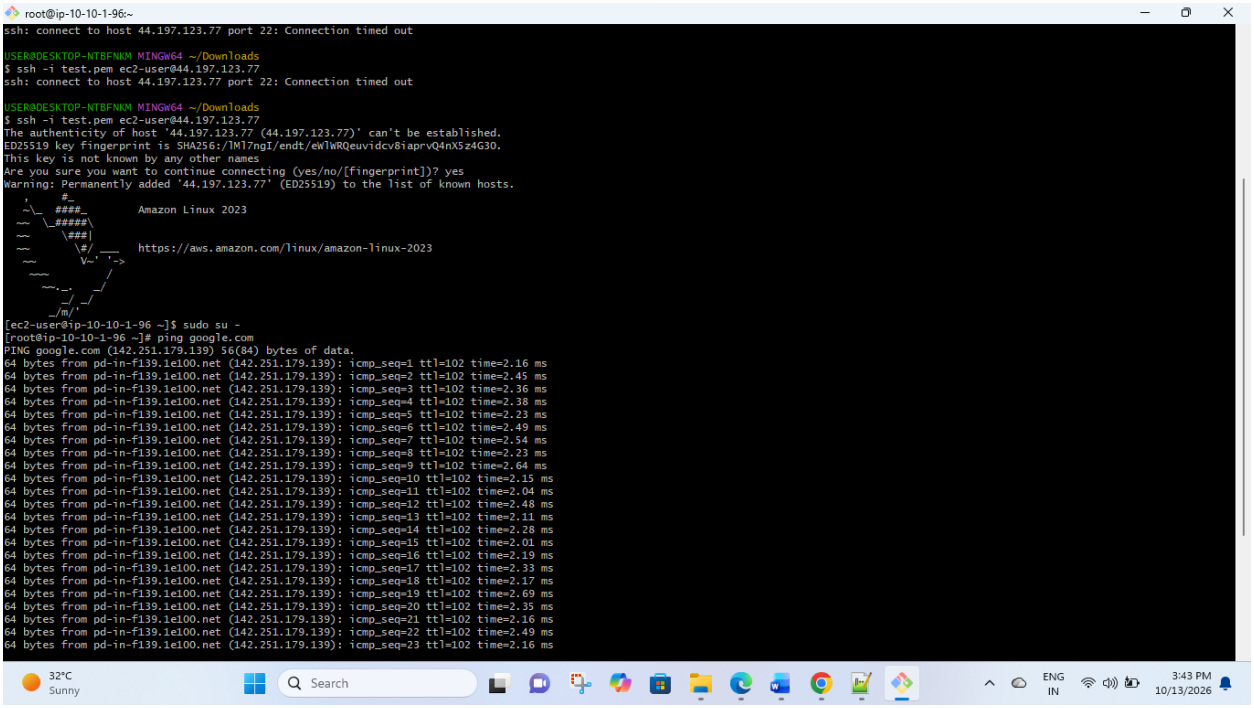


Successfully added internet gateway in routes added destination cidr .

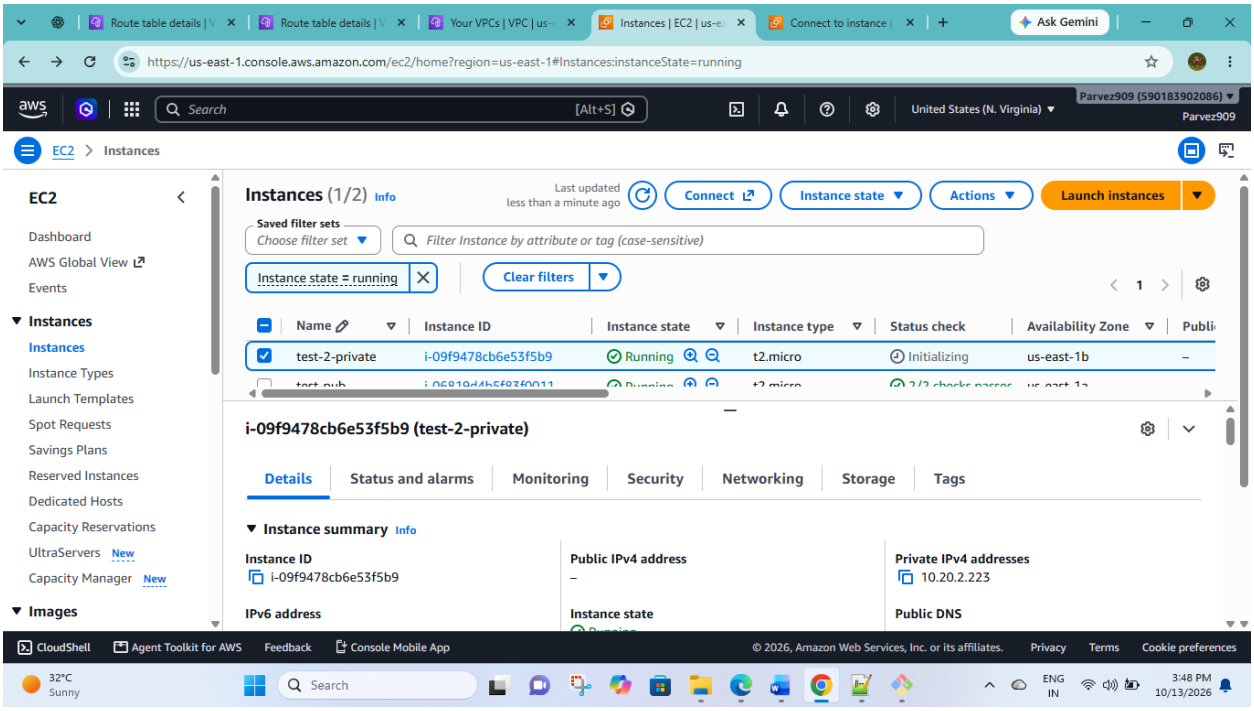
7 Launch EC2 instances

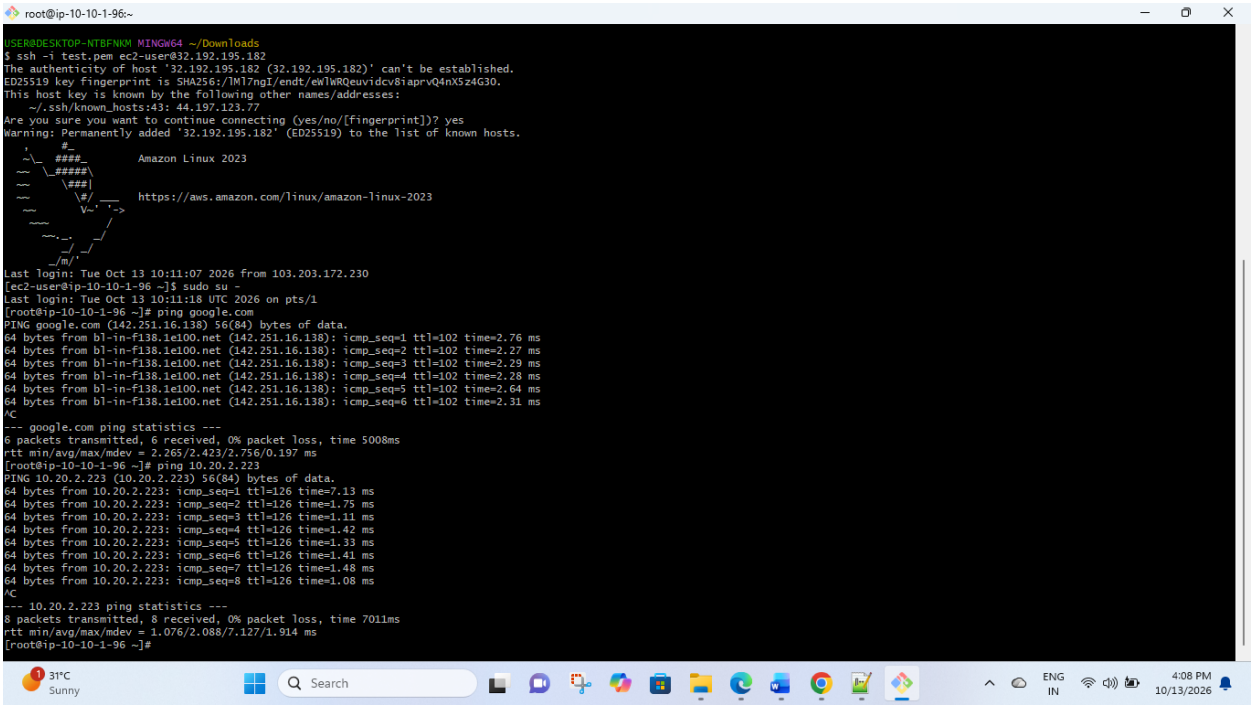




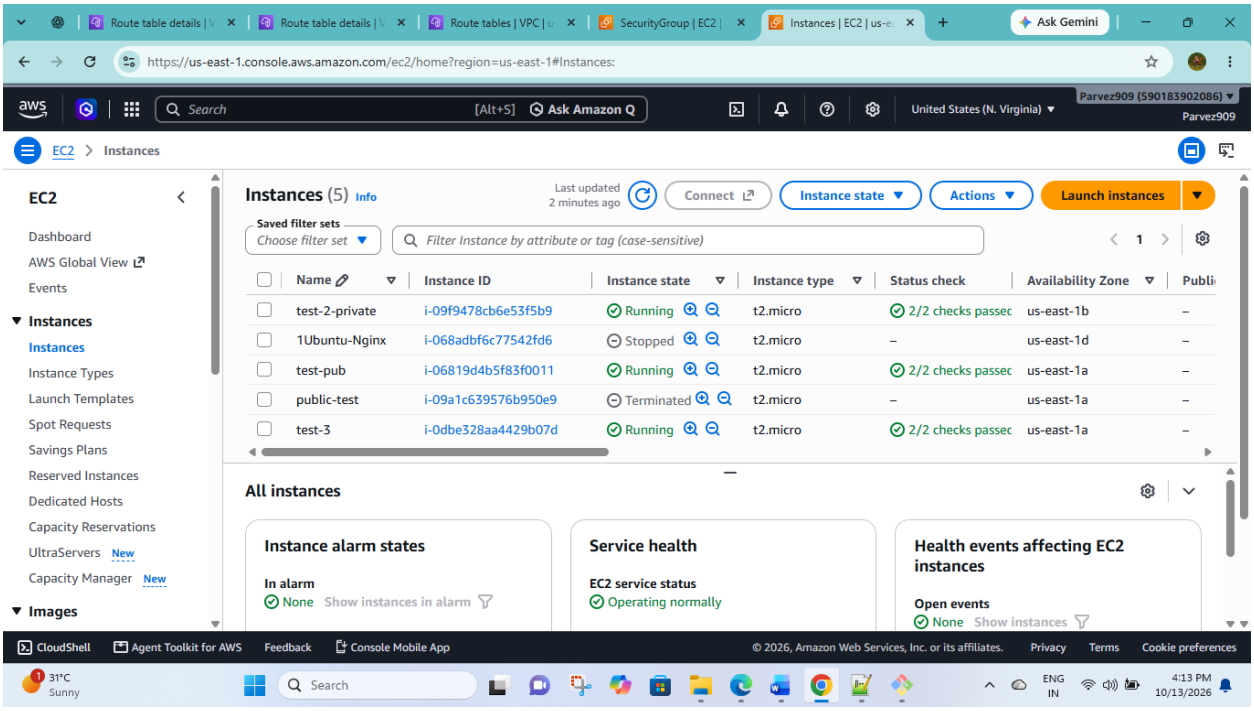


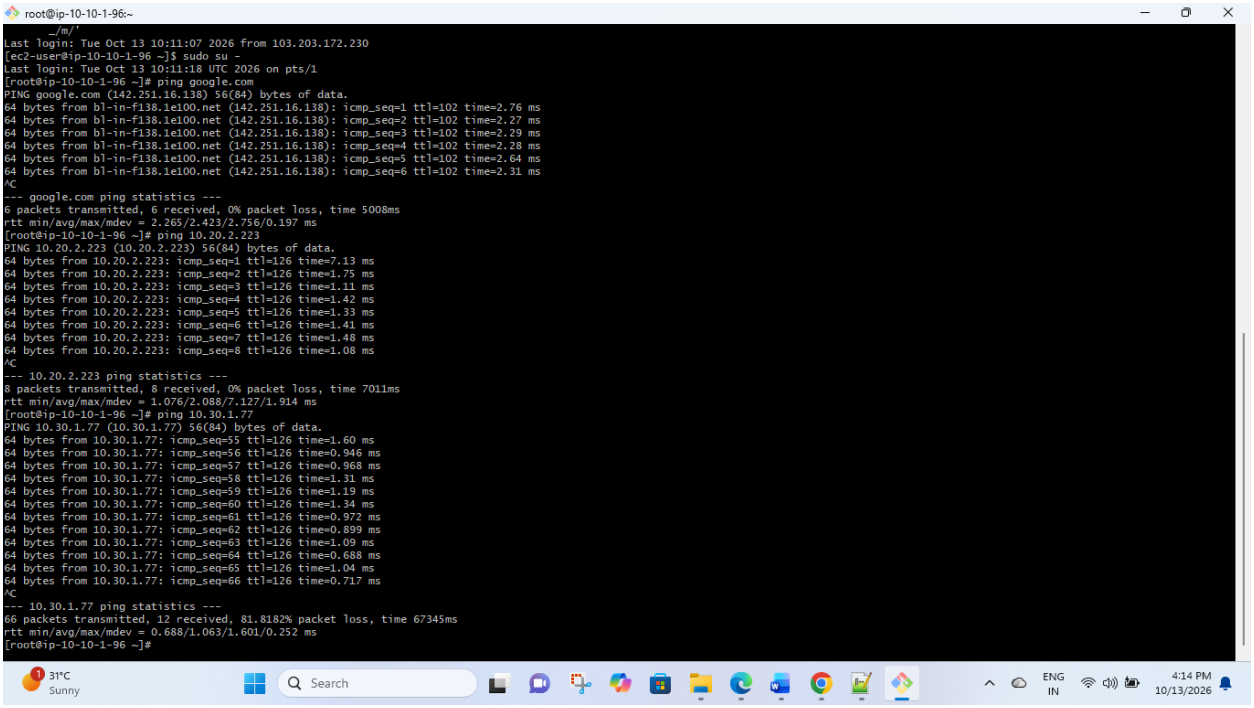
vpc2 Test private in


Route table details | V
Route table details | V
Your VPCs | VPC | us-e
Instances | EC2 | us-ea
Connect to instance
Ask Gemini
https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:instanceState=running
Search
[Alt+S]
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
Instances (1/2) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
test-2-private
i-09f9478cb6e53f5b9
Running
t2.micro
Initializing
us-east-1b
i-09f9478cb6e53f5b9 (test-2-private)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-09f9478cb6e53f5b9
Public IPv4 address
–
Private IPv4 addresses
10.20.2.223
IPv6 address
Instance state
Public DNS
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
32°C
Sunny
Search
ENG
IN
3:48 PM
10/13/2026

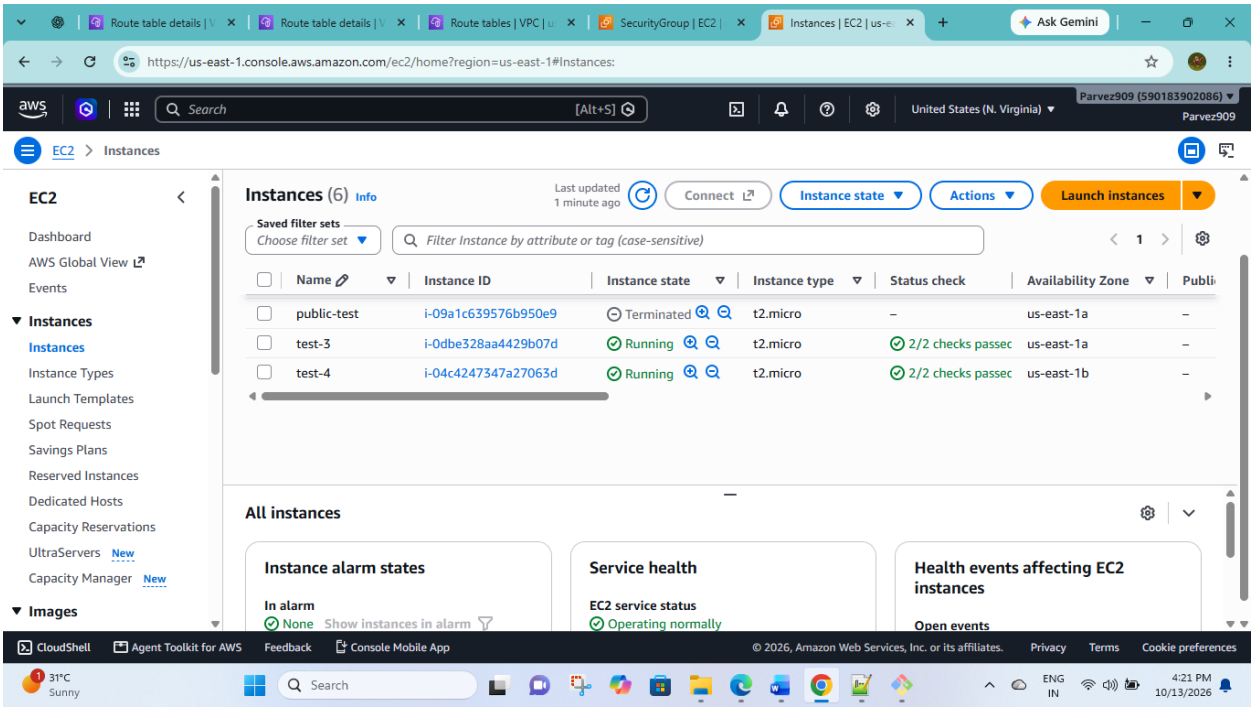

root@ip-10-10-1-96:~
USER@DESKTOP-NTBFNKM MINGW64 ~/Downloads
$ ssh -i test.pem ec2-user@32.192.195.182
The authenticity of host '32.192.195.182 (32.192.195.182)' can't be established.
ED25519 key fingerprint is SHA256:/lMl7ngI/endt/eWlWRQeuvidcv8iaprvQ4nX5z4G30.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:43: 44.197.123.77
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '32.192.195.182' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Tue Oct 13 10:11:07 2026 from 103.203.172.230
[ec2-user@ip-10-10-1-96 ~]$ sudo su -
Last login: Tue Oct 13 10:11:18 UTC 2026 on pts/1
[root@ip-10-10-1-96 ~]# ping google.com
PING google.com (142.251.16.138) 56(84) bytes of data.
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=1 ttl=102 time=2.76 ms
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=2 ttl=102 time=2.27 ms
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=3 ttl=102 time=2.29 ms
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=4 ttl=102 time=2.28 ms
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=5 ttl=102 time=2.64 ms
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=6 ttl=102 time=2.31 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 2.265/2.423/2.756/0.197 ms
[root@ip-10-10-1-96 ~]# ping 10.20.2.223
PING 10.20.2.223 (10.20.2.223) 56(84) bytes of data.
64 bytes from 10.20.2.223: icmp_seq=1 ttl=126 time=7.13 ms
64 bytes from 10.20.2.223: icmp_seq=2 ttl=126 time=1.75 ms
64 bytes from 10.20.2.223: icmp_seq=3 ttl=126 time=1.11 ms
64 bytes from 10.20.2.223: icmp_seq=4 ttl=126 time=1.42 ms
64 bytes from 10.20.2.223: icmp_seq=5 ttl=126 time=1.33 ms
64 bytes from 10.20.2.223: icmp_seq=6 ttl=126 time=1.41 ms
64 bytes from 10.20.2.223: icmp_seq=7 ttl=126 time=1.48 ms
64 bytes from 10.20.2.223: icmp_seq=8 ttl=126 time=1.08 ms
^C
--- 10.20.2.223 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7011ms
rtt min/avg/max/mdev = 1.076/2.088/7.127/1.914 ms
[root@ip-10-10-1-96 ~]#
31°C
Sunny
Search
ENG
IN
4:08 PM
10/13/2026

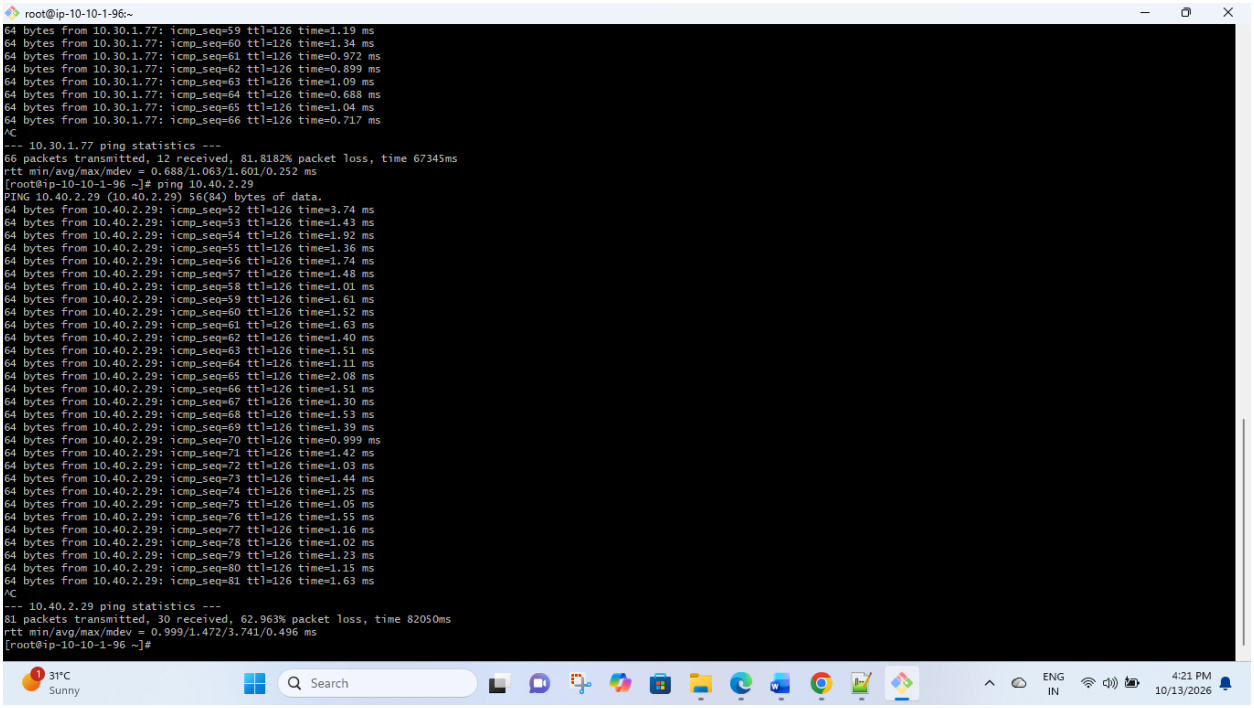
Test3


Route table details | V
Route table details | V
Route tables | VPC | u
SecurityGroup | EC2 |
Instances | EC2 | us-e
Ask Gemini
https://us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
Instances (5) Info
Last updated 2 minutes ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
test-2-private i-09f9478cb6e53f5b9 Running t2.micro 2/2 checks passec us-east-1b
1Ubuntu-Nginx i-068adbf6c77542fd6 Stopped t2.micro – us-east-1d
test-pub i-06819d4b5f83f0011 Running t2.micro 2/2 checks passec us-east-1a
public-test i-09a1c639576b950e9 Terminated t2.micro – us-east-1a
test-3 i-0dbe328aa4429b07d Running t2.micro 2/2 checks passec us-east-1a
All instances
Instance alarm states
In alarm
None Show instances in alarm
Service health
EC2 service status
Operating normally
Health events affecting EC2 instances
Open events
None Show instances
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
31°C Sunny
Search
ENG IN
4:13 PM 10/13/2026


root@ip-10-10-1-96:~
Last login: Tue Oct 13 10:11:07 2026 from 103.203.172.230
[ec2-user@ip-10-10-1-96 ~]$ sudo su -
Last login: Tue Oct 13 10:11:18 UTC 2026 on pts/1
[root@ip-10-10-1-96 ~]# ping google.com
PING google.com (142.251.16.138) 56(84) bytes of data.
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=1 ttl=102 time=2.76 ms
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=2 ttl=102 time=2.27 ms
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=3 ttl=102 time=2.29 ms
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=4 ttl=102 time=2.28 ms
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=5 ttl=102 time=2.64 ms
64 bytes from bl-in-f138.1e100.net (142.251.16.138): icmp_seq=6 ttl=102 time=2.31 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 2.265/2.423/2.756/0.197 ms
[root@ip-10-10-1-96 ~]# ping 10.20.2.223
PING 10.20.2.223 (10.20.2.223) 56(84) bytes of data.
64 bytes from 10.20.2.223: icmp_seq=1 ttl=126 time=7.13 ms
64 bytes from 10.20.2.223: icmp_seq=2 ttl=126 time=1.75 ms
64 bytes from 10.20.2.223: icmp_seq=3 ttl=126 time=1.11 ms
64 bytes from 10.20.2.223: icmp_seq=4 ttl=126 time=1.42 ms
64 bytes from 10.20.2.223: icmp_seq=5 ttl=126 time=1.33 ms
64 bytes from 10.20.2.223: icmp_seq=6 ttl=126 time=1.41 ms
64 bytes from 10.20.2.223: icmp_seq=7 ttl=126 time=1.48 ms
64 bytes from 10.20.2.223: icmp_seq=8 ttl=126 time=1.08 ms
^C
--- 10.20.2.223 ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7011ms
rtt min/avg/max/mdev = 1.076/2.088/7.127/1.914 ms
[root@ip-10-10-1-96 ~]# ping 10.30.1.77
PING 10.30.1.77 (10.30.1.77) 56(84) bytes of data.
64 bytes from 10.30.1.77: icmp_seq=55 ttl=126 time=1.60 ms
64 bytes from 10.30.1.77: icmp_seq=56 ttl=126 time=0.946 ms
64 bytes from 10.30.1.77: icmp_seq=57 ttl=126 time=0.968 ms
64 bytes from 10.30.1.77: icmp_seq=58 ttl=126 time=1.31 ms
64 bytes from 10.30.1.77: icmp_seq=59 ttl=126 time=1.19 ms
64 bytes from 10.30.1.77: icmp_seq=60 ttl=126 time=1.34 ms
64 bytes from 10.30.1.77: icmp_seq=61 ttl=126 time=0.972 ms
64 bytes from 10.30.1.77: icmp_seq=62 ttl=126 time=0.899 ms
64 bytes from 10.30.1.77: icmp_seq=63 ttl=126 time=1.09 ms
64 bytes from 10.30.1.77: icmp_seq=64 ttl=126 time=0.688 ms
64 bytes from 10.30.1.77: icmp_seq=65 ttl=126 time=1.04 ms
64 bytes from 10.30.1.77: icmp_seq=66 ttl=126 time=0.717 ms
^C
--- 10.30.1.77 ping statistics ---
66 packets transmitted, 12 received, 81.8182% packet loss, time 67345ms
rtt min/avg/max/mdev = 0.688/1.063/1.601/0.252 ms
[root@ip-10-10-1-96 ~]#
31°C Sunny
Search
ENG IN
4:14 PM 10/13/2026

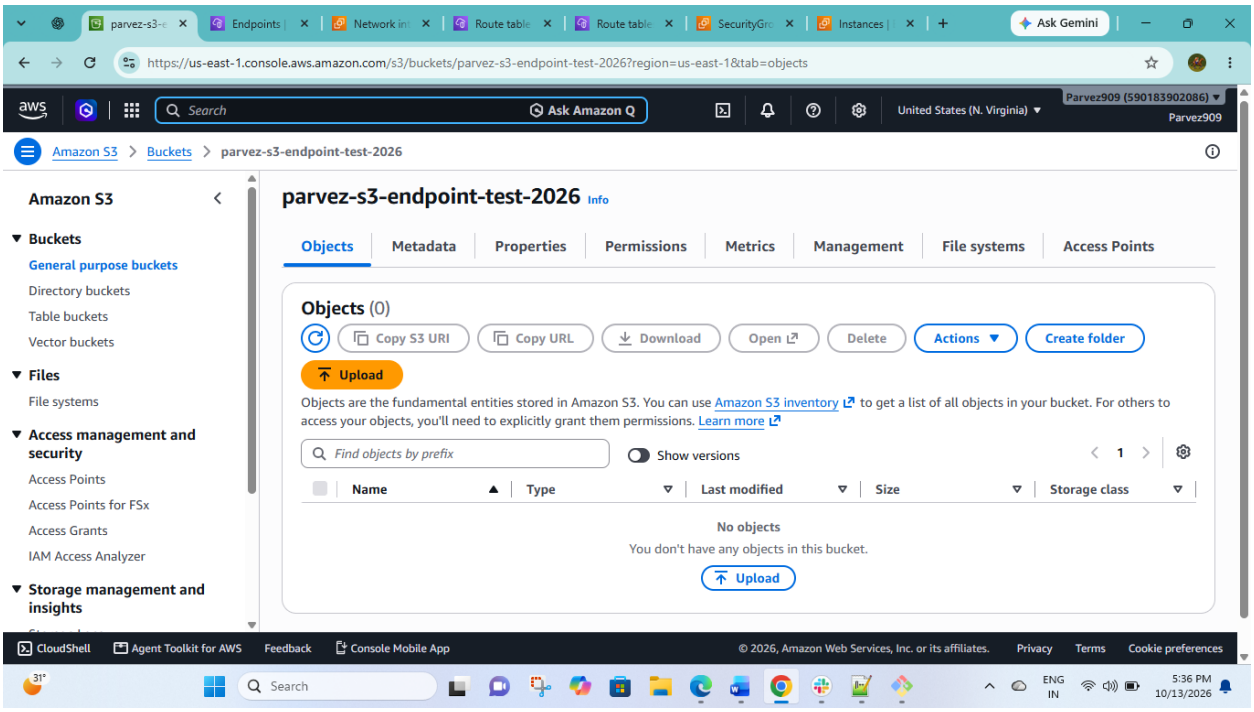
Test4



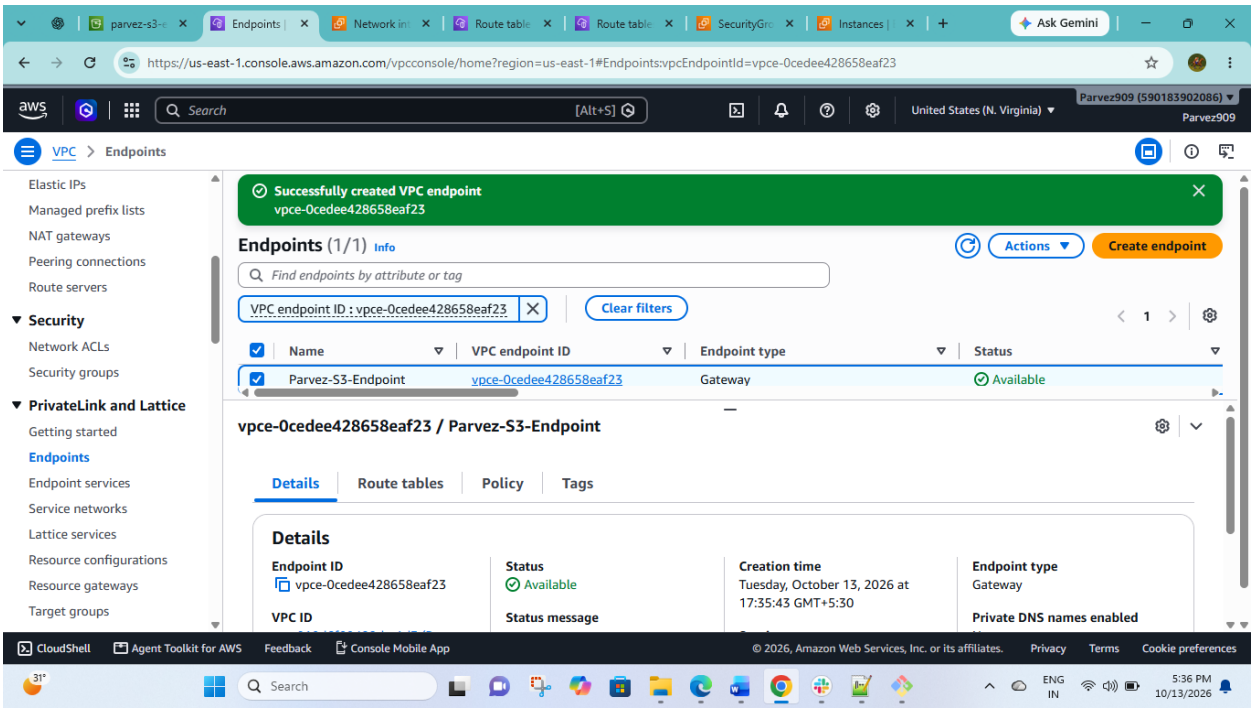


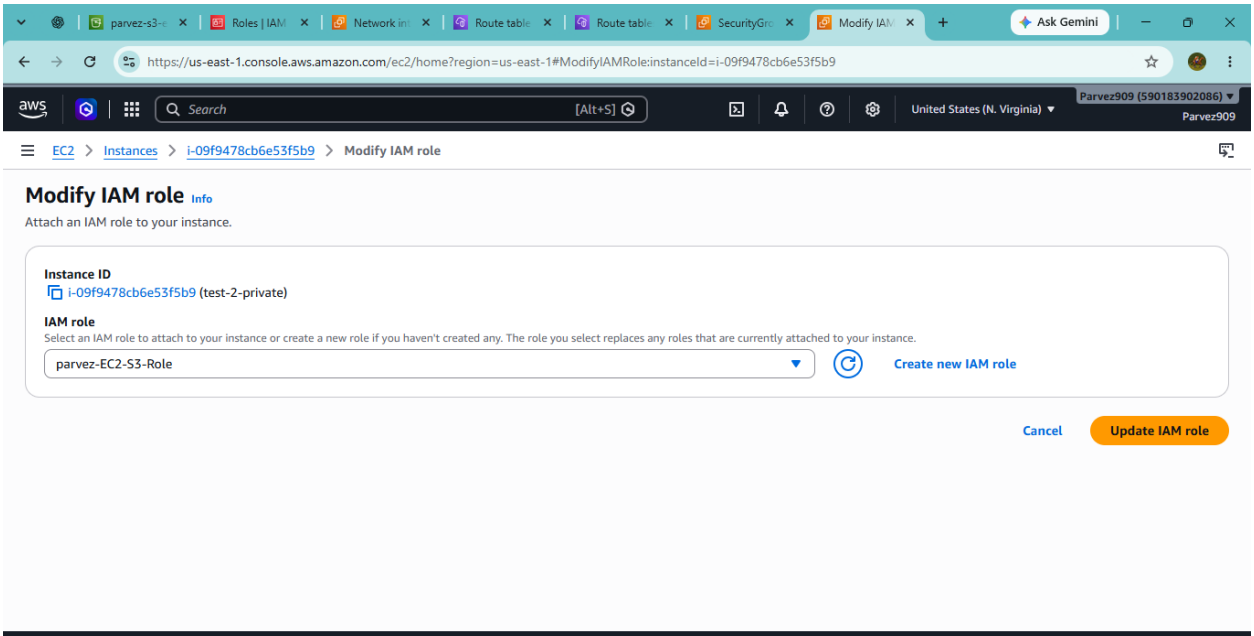
2. Configure VPC endpoints to securely access AWS services without Internet Gateways or NAT Gateways.

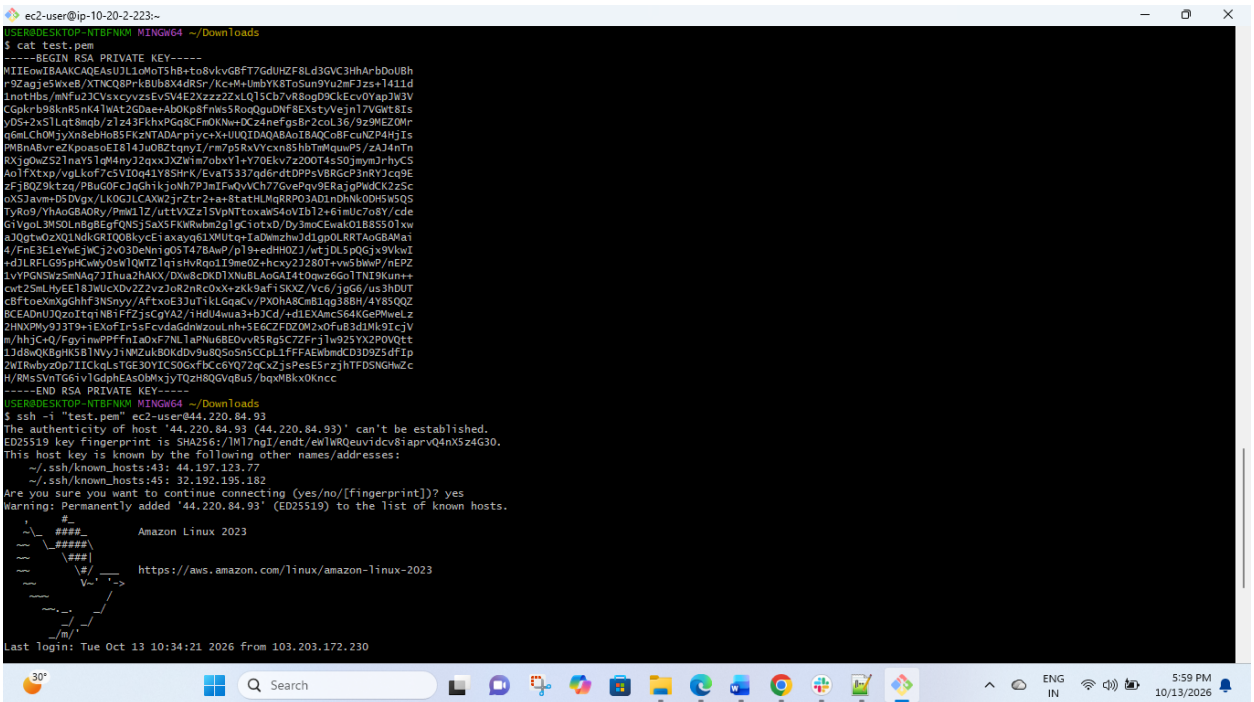
Create s3

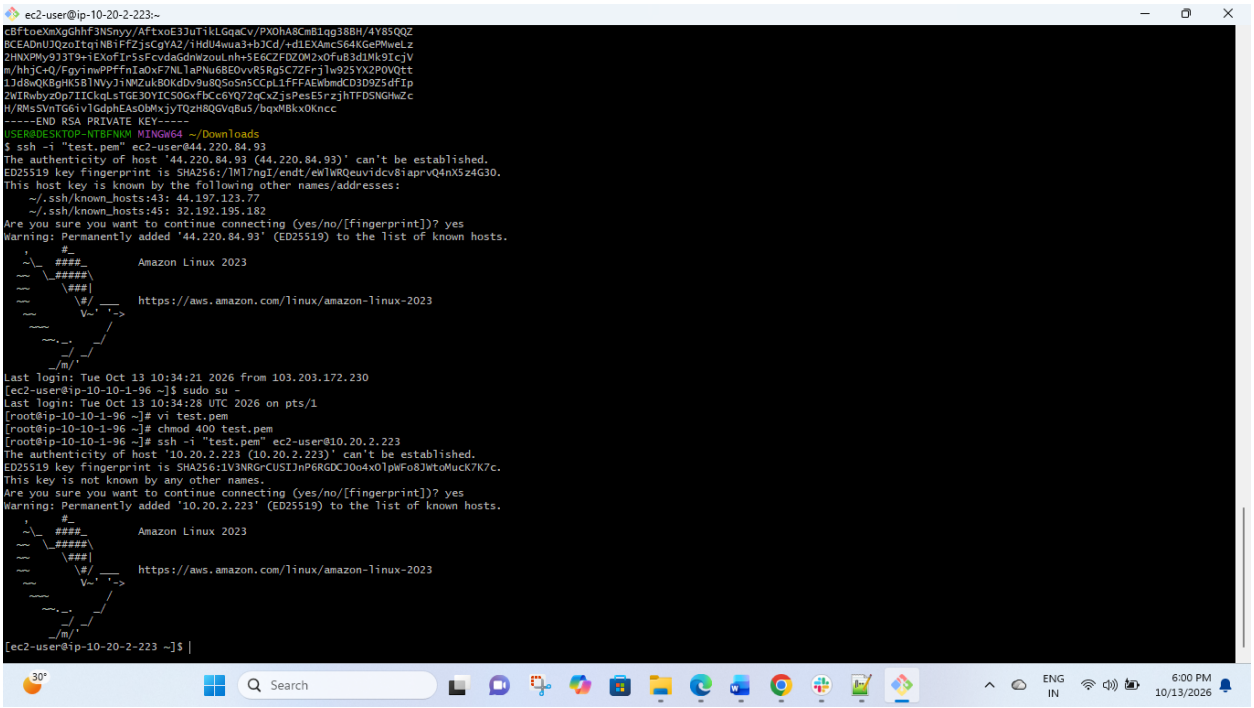


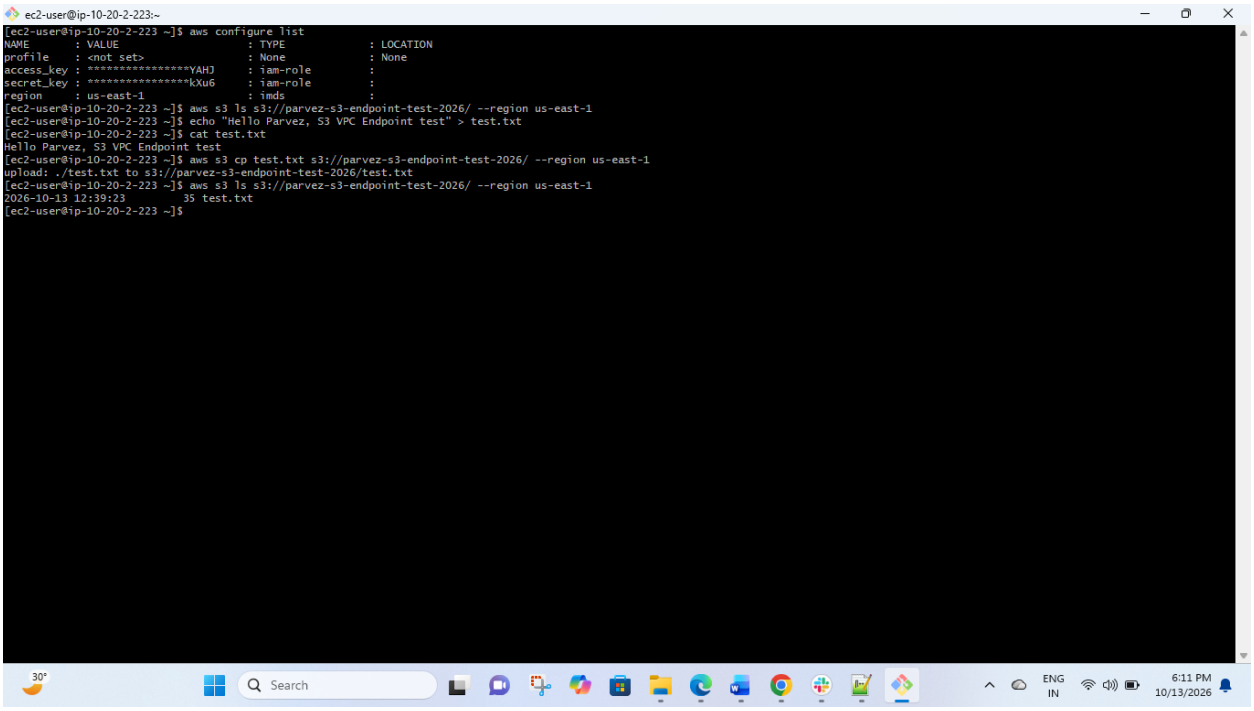
Create endpoint.

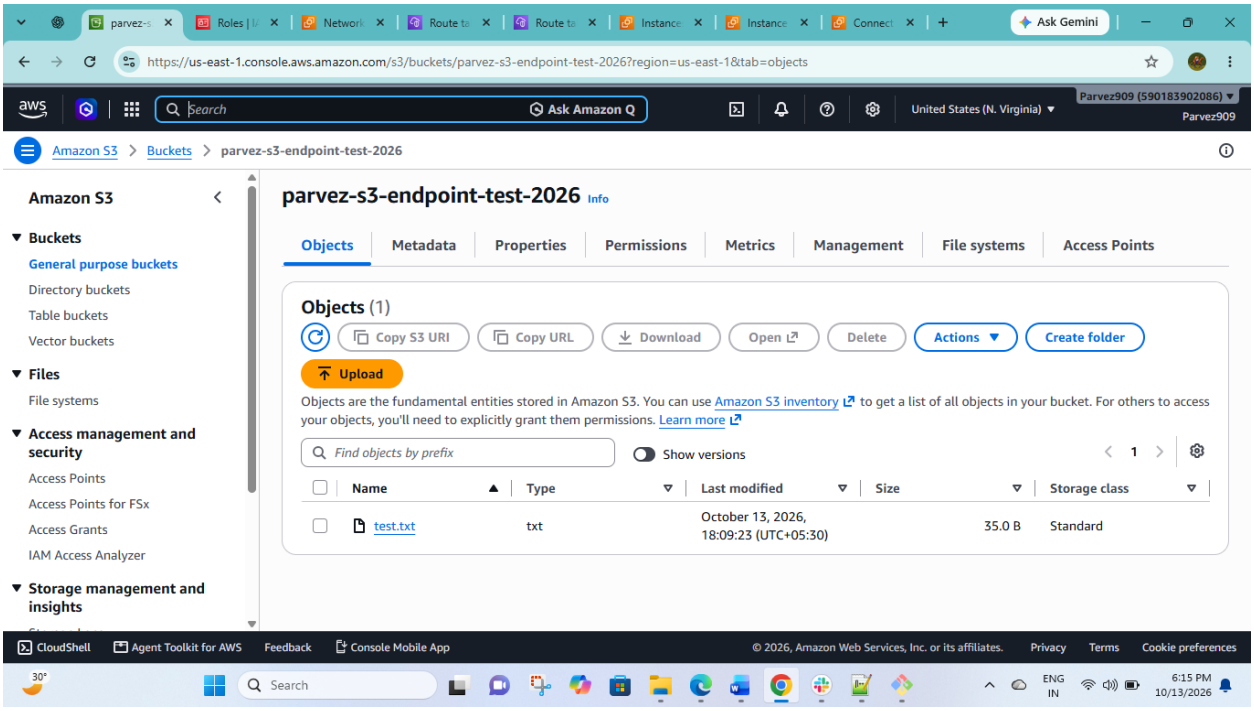



Modify IAM role
Info
Attach an IAM role to your instance.
Instance ID
i-09f9478cb6e53f5b9 (test-2-private)
IAM role
Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.
parvez-EC2-S3-Role
Create new IAM role
Cancel
Update IAM role


ec2-user@ip-10-20-2-223:~
USER@DESKTOP-NTBFNKM MINGW64 ~/Downloads
$ cat test.pem
-----BEGIN RSA PRIVATE KEY-----
-----END RSA PRIVATE KEY-----
USER@DESKTOP-NTBFNKM MINGW64 ~/Downloads
$ ssh -i "test.pem" ec2-user@44.220.84.93
The authenticity of host '44.220.84.93 (44.220.84.93)' can't be established.
ED25519 key fingerprint is SHA256:lMl7ngI/endt/eWlWRQeuvidcv8iaprvQ4nX524G30.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:43: 44.197.123.77
~/.ssh/known_hosts:45: 32.192.195.182
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.220.84.93' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Tue Oct 13 10:34:21 2026 from 103.203.172.230


ec2-user@ip-10-20-2-223:~
cBftoeXmXgGhhf3NSnyy/AftxoE3JuTikLGqaCv/PXOhA8CmB1qg38BH/4Y85QQZ
BCEADnUJQzoItqiNBiFfZjsCgYA2/iHdU4wua3+bJCd/+d1EXAmcS64KGePMweLz
2HNXPMy9J3T9+iEXofIr5sFcvdaGdnWzouLnh+SE6CZFDZOM2xOfuB3d1Mk9IcjV
m/hhjC+Q/FgyinwPPffnIaOxF7NLlaPNu6BEOvvR5Rg5C7ZFrjlw92SYX2POVQtt
1Jd8wQK8gHK5BlNVyJiNMZukBOKdDv9u8QSoSn5CCpL1fFFAEWbmdCD3D9Z5dfIp
2WIRwbyzOp7IICkqLsTGE3OYICSOGxfbCc6YQ72qCxZjsPesE5rzjhTFDSNGHwZc
H/RMsSVnTG6ivlGdphEAsObMxjyTQzH8QGVqBu5/bqxMBkxOKncc
-----END RSA PRIVATE KEY-----
USER@DESKTOP-NTBFNKM MINGW64 ~/Downloads
$ ssh -i "test.pem" ec2-user@44.220.84.93
The authenticity of host '44.220.84.93 (44.220.84.93)' can't be established.
ED25519 key fingerprint is SHA256:/lMl7ngI/endt/eWlWRQeuvidcv8iaprvQ4nX5z4G3O.
This host key is known by the following other names/addresses:
~/.ssh/known_hosts:43: 44.197.123.77
~/.ssh/known_hosts:45: 32.192.195.182
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '44.220.84.93' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
Last login: Tue Oct 13 10:34:21 2026 from 103.203.172.230
[ec2-user@ip-10-10-1-96 ~]$ sudo su -
Last login: Tue Oct 13 10:34:28 UTC 2026 on pts/1
[root@ip-10-10-1-96 ~]# vi test.pem
[root@ip-10-10-1-96 ~]# chmod 400 test.pem
[root@ip-10-10-1-96 ~]# ssh -i "test.pem" ec2-user@10.20.2.223
The authenticity of host '10.20.2.223 (10.20.2.223)' can't be established.
ED25519 key fingerprint is SHA256:1V3NRGrCUSIJnP6RGDCJOo4xOlpWFo8JWtoMucK7K7c.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.2.223' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-20-2-223 ~]$
Search
ENG
IN
6:00 PM
10/13/2026


ec2-user@ip-10-20-2-223:~
[ec2-user@ip-10-20-2-223 ~]$ aws configure list
NAME : VALUE : TYPE : LOCATION
profile : <not set> : None : None
access_key : ****************YAHJ : iam-role :
secret_key : ****************kXu6 : iam-role :
region : us-east-1 : imds :
[ec2-user@ip-10-20-2-223 ~]$ aws s3 ls s3://parvez-s3-endpoint-test-2026/ --region us-east-1
[ec2-user@ip-10-20-2-223 ~]$ echo "Hello Parvez, S3 VPC Endpoint test" > test.txt
[ec2-user@ip-10-20-2-223 ~]$ cat test.txt
Hello Parvez, S3 VPC Endpoint test
[ec2-user@ip-10-20-2-223 ~]$ aws s3 cp test.txt s3://parvez-s3-endpoint-test-2026/ --region us-east-1
upload: ./test.txt to s3://parvez-s3-endpoint-test-2026/test.txt
[ec2-user@ip-10-20-2-223 ~]$ aws s3 ls s3://parvez-s3-endpoint-test-2026/ --region us-east-1
2026-10-13 12:39:23 35 test.txt
[ec2-user@ip-10-20-2-223 ~]$
Search
ENG
IN
6:11 PM
10/13/2026


https://us-east-1.console.aws.amazon.com/s3/buckets/parvez-s3-endpoint-test-2026?region=us-east-1&tab=objects
Search
Ask Amazon Q
United States (N. Virginia)
Parvez909 (590183902086)
Parvez909
Amazon S3 > Buckets > parvez-s3-endpoint-test-2026
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
parvez-s3-endpoint-test-2026 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Objects (1)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Show versions
Name
Type
Last modified
Size
Storage class
test.txt
txt
October 13, 2026, 18:09:23 (UTC+05:30)
35.0 B
Standard
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
Search
ENG IN
6:15 PM 10/13/2026